INCH-POUND
MIL-I-24092/5A
21 September 1993
SUPERSEDING
MIL-I-24092/5
4 May 1989

MILITARY SPECIFICATION SHEET

INSULATING RESIN, SOLVENTLESS, BAKING, FLEXIBLE, FOR DIP PROCESSING, GRADE SF, CLASS 130 TO 180 THERMAL CLASS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24092.

REQUIREMENTS:

1. Grade - SF. Clear, solventless flexible baking resin.

2. Class - 130 to 180. The cured resin shall meet the requirements of the specified thermal class when determined in accordance with MIL-I-24092.

3. Composition - IV. Composition IV consists of resins that are in solution with either an organic monomer or an organic diluent. Such materials are not normally considered solvents since they are reactive and become part of the cured polymer.

4. The resin shall meet the qualification and quality conformance requirements specified in tables I and II herein when tested in accordance with MIL-I-24092.

AMSC N/A FSC 5970

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:01Z
Check the source to verify that this is the current version before use.

TABLE I. Qualification requirements, grade SF

Characteristics (tests)	Units	Requirements
Catalyzed storage life	Months, minimum	6.0
Flash point	°C, minimum (military use only)	93
Viscosity at 20 r/min	Centipoise	150 to 1300
Build, as received	Mil, minimum	0.3
Dielectric strength:		
96/23/50	Volts per mil, minimum	1500
96/23/96	Minimum	1125
24/23/water	Minimum	1125
Bond strength:		
at 23°C	Pounds force, minimum	10
at 150°C	Pounds force, minimum	1
Thermal endurance (M2, R2) only:		
Class 130	Temperature index, °C, minimum	130
Class 155	Temperature index, °C, minimum	155
Class 180	Temperature index, °C, minimum	180
Salt water proofness	Hours, minimum	100
Hydrolytic stability	Percent decrease in bond strength, maximum	To be determined
Chemical resistance:		
Hydraulic fluid	Percent decrease in bond strength, maximum	50
Lubrication oil		50
Cleaning fluid		50
Distilled water		50
Detergent solution		50

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:01Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance requirements, grade SF.

Characteristics (tests)	Units	Requirements
Gel time	Percent of as-received value	±10
Viscosity at 20 r/min	Centipoise, maximum percent of qualification value	±25
Build, as received	Mil, minimum	0.3
Specific gravity	Percent of as-received value	±10
Bond strength at 23°C	Pounds force, minimum	10

NOTE:

Part or Identifying Number (PIN). Part number shall be M24092/5-(dash number indicating thermal class). For example, M24092/5-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance. The PIN shall be the same format as specified in MIL-I-24092.

Custodians:
Army - ER
Navy - SH
Air Force - 11

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - ME
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-1078)

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:01Z
Check the source to verify that this is the current version before use.

shall be provided on four of the wires contained within the overjacket specified below. Unless